

Dunnett's Critical Value Table

Complete Reference Card — All Alpha Levels + How-To-Read Guide

Two-Tailed: $\alpha = 0.10, 0.05, 0.01$ | One-Tailed: $\alpha = 0.05, 0.01$

$k = 2$ to 9 treatment groups | Error $df = 2$ to ∞

StatisticsFundamentals.com/tables/dunnetts-table/

How to Use Dunnett's Table

- 1
- Run a one-way ANOVA. If F is significant at your chosen alpha, proceed. Record MSE (Mean Square Error) and df_{error} from the ANOVA summary table.
- 2
- Count k — the number of treatment groups. Do NOT count the control group. Example: 3 drug doses + 1 placebo control $\rightarrow k = 3$.
- 3
- Note df_{error} from ANOVA. $df_{\text{error}} = N - (k + 1)$, where N = total observations. This value determines the row in Dunnett's Table.
- 4
- Choose alpha and tail direction. Use $\alpha = 0.05$ for standard research; $\alpha = 0.01$ for strict testing. Use two-tailed unless you have a directional hypothesis.
- 5
- Look up d^* (critical value). Find the row for df_{error} and the column for k . The intersection is your critical value d^* .
- 6
- Compute d and decide. For each treatment i : $d = (\text{mean}_i - \text{mean}_0) / \text{SE}$, where $\text{SE} = \sqrt{\text{MSE} * (1/n_i + 1/n_0)}$. If $|d| > d^*$, that treatment differs significantly from control.

Key Formulas

Test Statistic	Error Degrees of Freedom
$d = (\text{mean}_i - \text{mean}_0) / SE$	$df_{\text{error}} = N - (k + 1)$
Standard Error	Decision Rule
$SE = \sqrt{MSE \times (1/n_i + 1/n_0)}$	Reject H0 if $ d > d^*$

Quick Lookup: Most Commonly Used Critical Values

Two-tailed Dunnett's d* for the most frequently encountered combinations of df_error and k. Yellow row (df=20) is the most common in published research. Full tables covering df = 2–∞ and k = 2–9 appear on the following pages.

df_error	k=2 (α=0.05)	k=3 (α=0.05)	k=4 (α=0.05)	k=2 (α=0.01)	k=3 (α=0.01)	k=4 (α=0.01)
5	2.57	2.91	3.11	3.70	4.15	4.43
10	2.34	2.60	2.76	2.57	2.81	2.95
15	2.24	2.50	2.62	2.53	2.75	2.89
20	2.08	2.34	2.49	2.51	2.73	2.86
30	2.05	2.31	2.46	2.48	2.70	2.83
60	2.02	2.28	2.42	2.45	2.66	2.79
120	2.01	2.27	2.41	2.44	2.64	2.77
∞	2.00	2.26	2.40	2.42	2.63	2.75

Source: Dunnett (1955) JASA; Kutner et al. (2005). All two-tailed values.

Dunnett's Test vs Other Post-Hoc Procedures — When to Use Which

Feature	Dunnett's	Tukey's HSD	Bonferroni	Scheffe
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Comparisons made	Each treatment vs control only	All pairs	Any pre-specified	All contrasts
Best use case	Treatment vs one control	All-pair comparisons	Few planned tests	Complex contrasts
Power (control comps)	HIGHEST	Lower	Lower	Lowest
FWER controlled	Yes (at alpha)	Yes (at alpha)	Yes (conservative)	Yes (conservative)
Requires ANOVA	Yes	Yes	Not strictly	Yes

Dunnett's test is the most powerful post-hoc test when comparisons are restricted to treatments vs a single control. Using Tukey's HSD in this situation wastes power.

Worked Example: 3 Drug Doses vs Placebo Control

Scenario: A pharmacologist tests Low, Medium, and High doses of a drug (k=3) against a placebo control (n=6 per group, N=24). One-way ANOVA gives MSE=12.00, df_error=20, F=5.14 (significant at alpha=0.05). Applying Dunnett's two-tailed test at alpha=0.05: $d^* = 2.54$ (from table: k=3, df=20). $SE = \sqrt{12.00 \times (1/6+1/6)} = 2.00$. Low dose: d=1.25 (not significant). Medium dose: d=2.75 > 2.54 (significant*). High dose: d=4.00 > 2.54 (significant**). Conclusion: Medium and High dose significantly reduce blood pressure vs placebo; Low dose does not. FWER controlled at 5% across all comparisons.

Dunnett's Critical Values — Two-Tailed $\alpha = 0.05$ (Standard — 95% Confidence)								
Most commonly used in academic research, experiments, and homework Yellow row = $df = 20$ (most common)								
df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
2	2.57	3.03	3.29	3.48	3.62	3.73	3.82	3.90
3	2.34	2.68	2.88	3.04	3.16	3.26	3.34	3.41
4	2.25	2.56	2.74	2.88	3.00	3.09	3.17	3.24
5	2.20	2.49	2.67	2.80	2.91	3.00	3.07	3.14
6	2.17	2.45	2.62	2.75	2.86	2.94	3.01	3.08
7	2.15	2.43	2.59	2.72	2.82	2.91	2.97	3.04
8	2.13	2.41	2.57	2.70	2.79	2.88	2.94	3.01
9	2.12	2.40	2.55	2.68	2.77	2.86	2.92	2.99
10	2.11	2.38	2.54	2.66	2.76	2.84	2.91	2.97
11	2.10	2.37	2.53	2.65	2.74	2.83	2.89	2.96
12	2.10	2.37	2.52	2.64	2.73	2.82	2.88	2.95
13	2.09	2.36	2.51	2.63	2.73	2.81	2.87	2.94
14	2.09	2.36	2.51	2.63	2.72	2.80	2.86	2.93
15	2.09	2.35	2.50	2.62	2.72	2.80	2.86	2.92
16	2.08	2.35	2.50	2.62	2.71	2.79	2.85	2.92
17	2.08	2.35	2.50	2.61	2.71	2.79	2.85	2.91
18	2.08	2.34	2.49	2.61	2.70	2.78	2.84	2.91
19	2.08	2.34	2.49	2.61	2.70	2.78	2.84	2.91
20	2.08	2.34	2.49	2.61	2.70	2.78	2.84	2.90
24	2.06	2.33	2.47	2.59	2.68	2.76	2.82	2.88

df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
30	2.05	2.31	2.46	2.57	2.66	2.74	2.80	2.86
40	2.04	2.30	2.44	2.56	2.64	2.72	2.78	2.84
60	2.02	2.28	2.42	2.53	2.62	2.70	2.76	2.81
120	2.01	2.27	2.41	2.52	2.61	2.68	2.74	2.80
∞	2.00	2.26	2.40	2.51	2.59	2.67	2.73	2.78

Source: Dunnett (1955) JASA 50:1096-1121. Two-Tailed $\alpha=0.05$. Visit: statisticsfundamentals.com/tables/dunnetts-table/

Dunnett's Critical Values — Two-Tailed $\alpha = 0.01$ (Strict — 99% Confidence)								
Use for strict significance control in clinical or regulatory settings Yellow row = df = 20 (most common)								
df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
2	3.70	4.15	4.43	4.63	4.79	4.92	5.03	5.12
3	3.09	3.43	3.64	3.80	3.93	4.03	4.12	4.20
4	2.88	3.17	3.35	3.50	3.62	3.71	3.79	3.87
5	2.76	3.03	3.20	3.33	3.44	3.53	3.61	3.68
6	2.70	2.95	3.11	3.24	3.34	3.43	3.50	3.57
7	2.65	2.90	3.05	3.18	3.27	3.36	3.43	3.50
8	2.62	2.86	3.01	3.13	3.23	3.31	3.38	3.45
9	2.59	2.83	2.98	3.10	3.19	3.28	3.35	3.41
10	2.57	2.81	2.95	3.07	3.16	3.24	3.32	3.38
11	2.56	2.79	2.93	3.05	3.14	3.22	3.29	3.36
12	2.55	2.78	2.92	3.03	3.13	3.21	3.28	3.34
13	2.54	2.77	2.91	3.02	3.11	3.19	3.26	3.33
14	2.53	2.76	2.90	3.01	3.10	3.18	3.25	3.31

df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
15	2.53	2.75	2.89	3.00	3.09	3.17	3.24	3.30
16	2.52	2.75	2.88	2.99	3.08	3.16	3.23	3.29
17	2.52	2.74	2.88	2.99	3.07	3.15	3.22	3.28
18	2.51	2.74	2.87	2.98	3.07	3.14	3.21	3.28
19	2.51	2.73	2.87	2.97	3.06	3.14	3.21	3.27
20	2.51	2.73	2.86	2.97	3.06	3.13	3.20	3.27
24	2.50	2.71	2.85	2.95	3.04	3.11	3.18	3.24
30	2.48	2.70	2.83	2.93	3.02	3.09	3.16	3.22
40	2.47	2.68	2.81	2.92	3.00	3.07	3.14	3.20
60	2.45	2.66	2.79	2.89	2.97	3.05	3.11	3.17
120	2.44	2.64	2.77	2.87	2.95	3.02	3.09	3.15
∞	2.42	2.63	2.75	2.85	2.93	3.00	3.07	3.12

Source: Dunnett (1955) JASA 50:1096-1121. Two-Tailed $\alpha=0.01$. Visit: statisticsfundamentals.com/tables/dunnetts-table/

Dunnett's Critical Values — Two-Tailed | $\alpha = 0.10$ (Liberal — 90% Confidence)

Use for exploratory research or when Type II error is more costly than Type I | Yellow row = df = 20 (most common)

df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
2	2.03	2.44	2.68	2.86	2.99	3.10	3.19	3.27
3	1.91	2.25	2.46	2.61	2.73	2.83	2.91	2.98
4	1.85	2.17	2.36	2.50	2.61	2.70	2.78	2.85
5	1.81	2.12	2.30	2.44	2.54	2.63	2.71	2.77
6	1.78	2.09	2.26	2.40	2.50	2.59	2.66	2.73
7	1.77	2.07	2.24	2.37	2.47	2.55	2.63	2.69

df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
8	1.76	2.05	2.22	2.35	2.45	2.53	2.61	2.67
9	1.75	2.04	2.20	2.33	2.43	2.51	2.59	2.65
10	1.74	2.03	2.19	2.32	2.42	2.50	2.57	2.64
11	1.73	2.02	2.18	2.31	2.40	2.49	2.56	2.62
12	1.73	2.01	2.17	2.30	2.39	2.48	2.55	2.61
13	1.73	2.01	2.17	2.29	2.39	2.47	2.54	2.60
14	1.72	2.00	2.16	2.29	2.38	2.46	2.53	2.59
15	1.72	2.00	2.16	2.28	2.38	2.46	2.53	2.59
16	1.72	2.00	2.16	2.28	2.37	2.45	2.52	2.58
17	1.72	1.99	2.15	2.27	2.37	2.45	2.52	2.58
18	1.71	1.99	2.15	2.27	2.36	2.44	2.51	2.57
19	1.71	1.99	2.15	2.27	2.36	2.44	2.51	2.57
20	1.71	1.99	2.15	2.27	2.36	2.44	2.51	2.57
24	1.71	1.98	2.14	2.25	2.35	2.43	2.49	2.55
30	1.70	1.97	2.12	2.24	2.33	2.41	2.48	2.54
40	1.69	1.96	2.11	2.23	2.32	2.39	2.46	2.52
60	1.68	1.95	2.10	2.21	2.30	2.38	2.44	2.50
120	1.68	1.94	2.09	2.20	2.29	2.36	2.43	2.49
∞	1.67	1.93	2.08	2.19	2.28	2.35	2.42	2.48

Source: Dunnett (1955) JASA 50:1096-1121. Two-Tailed $\alpha=0.10$. Visit: statisticsfundamentals.com/tables/dunnetts-table/

Dunnett's Critical Values — One-Tailed | $\alpha = 0.05$ (Directional — 95% Confidence)

When all treatments are expected to differ from control in ONE direction only | Yellow row = df = 20 (most common)

df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
2	2.02	2.43	2.68	2.85	2.99	3.10	3.19	3.27
3	1.88	2.22	2.43	2.58	2.70	2.80	2.89	2.96
4	1.82	2.13	2.33	2.47	2.59	2.68	2.76	2.83
5	1.78	2.07	2.26	2.40	2.51	2.60	2.68	2.75
6	1.76	2.04	2.23	2.36	2.47	2.56	2.63	2.70
7	1.74	2.02	2.20	2.33	2.44	2.52	2.60	2.66
8	1.73	2.01	2.18	2.31	2.41	2.50	2.57	2.64
9	1.72	1.99	2.17	2.30	2.40	2.48	2.55	2.62
10	1.71	1.98	2.16	2.28	2.38	2.47	2.54	2.60
11	1.71	1.97	2.15	2.27	2.37	2.45	2.53	2.59
12	1.70	1.97	2.14	2.27	2.36	2.45	2.52	2.58
13	1.70	1.96	2.14	2.26	2.36	2.44	2.51	2.57
14	1.70	1.96	2.13	2.25	2.35	2.43	2.50	2.57
15	1.70	1.96	2.13	2.25	2.35	2.43	2.50	2.56
16	1.69	1.95	2.12	2.24	2.34	2.42	2.49	2.56
17	1.69	1.95	2.12	2.24	2.34	2.42	2.49	2.55
18	1.69	1.95	2.12	2.24	2.33	2.41	2.48	2.55
19	1.69	1.95	2.12	2.24	2.33	2.41	2.48	2.54
20	1.69	1.95	2.12	2.24	2.33	2.41	2.48	2.54
24	1.68	1.94	2.11	2.22	2.32	2.40	2.47	2.53
30	1.68	1.93	2.10	2.21	2.31	2.38	2.45	2.51
40	1.68	1.92	2.09	2.20	2.30	2.37	2.44	2.50

df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
60	1.67	1.91	2.08	2.19	2.28	2.36	2.43	2.49
120	1.66	1.91	2.07	2.18	2.27	2.35	2.41	2.47
∞	1.65	1.90	2.06	2.17	2.26	2.34	2.40	2.46

Source: Dunnett (1955) JASA 50:1096-1121. One-Tailed $\alpha=0.05$. Visit: statisticsfundamentals.com/tables/dunnetts-table/

Dunnett's Critical Values — One-Tailed alpha = 0.01 (Directional Strict — 99% Confidence)								
Strict directional testing — pharmaceutical dose-response, regulatory submissions Yellow row = df = 20 (most common)								
df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
2	2.92	3.34	3.61	3.81	3.96	4.09	4.20	4.29
3	2.54	2.87	3.08	3.24	3.37	3.48	3.57	3.65
4	2.39	2.68	2.87	3.01	3.13	3.22	3.31	3.38
5	2.30	2.57	2.74	2.88	2.99	3.08	3.16	3.23
6	2.24	2.50	2.67	2.80	2.90	2.99	3.07	3.14
7	2.20	2.46	2.62	2.75	2.85	2.93	3.01	3.07
8	2.17	2.43	2.59	2.71	2.81	2.89	2.97	3.03
9	2.15	2.40	2.56	2.68	2.78	2.86	2.93	3.00
10	2.13	2.38	2.53	2.65	2.75	2.83	2.90	2.97
11	2.12	2.37	2.52	2.64	2.73	2.82	2.88	2.95
12	2.11	2.36	2.51	2.63	2.72	2.80	2.87	2.94
13	2.10	2.35	2.50	2.62	2.71	2.79	2.86	2.92
14	2.10	2.34	2.49	2.61	2.70	2.78	2.85	2.91
15	2.09	2.34	2.49	2.60	2.70	2.77	2.84	2.91
16	2.09	2.33	2.48	2.60	2.69	2.77	2.84	2.90

df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
17	2.08	2.33	2.48	2.59	2.68	2.76	2.83	2.89
18	2.08	2.33	2.47	2.59	2.68	2.76	2.82	2.89
19	2.08	2.32	2.47	2.58	2.67	2.75	2.82	2.88
20	2.07	2.32	2.47	2.58	2.67	2.75	2.81	2.88
24	2.06	2.31	2.45	2.56	2.65	2.73	2.79	2.86
30	2.05	2.29	2.43	2.54	2.63	2.71	2.77	2.83
40	2.04	2.28	2.42	2.53	2.61	2.69	2.75	2.81
60	2.02	2.26	2.40	2.50	2.59	2.66	2.73	2.78
120	2.01	2.25	2.38	2.49	2.57	2.65	2.71	2.77
∞	2.00	2.23	2.37	2.47	2.56	2.63	2.69	2.75

Source: Dunnett (1955) JASA 50:1096-1121. One-Tailed $\alpha=0.01$. Visit: statisticsfundamentals.com/tables/dunnetts-table/

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